CRM RETENTION LAB / FINAL REPORT

After the first purchase, who should receive which CRM treatment?

Repurchase diagnosis, a three-arm experiment, and customer policies under budget, contact, and side-effect constraints

Fictional D2C coffee retailer · Marketing planning portfolio

+282 KRW
Incremental contribution profit per eligible customer

28.1%
Contact rate · 30% cap

15,363
Independent final-evaluation customers

Final decision: Modify or collect more evidence

Budget, contact, and both side-effect criteria are met. However, the 95% interval for incremental contribution profit is -154 to +640 KRW, so evidence is insufficient to establish a profit improvement.

2026.11.08 · Synthetic-data analysis
Training 15,284 / validation 15,167 / final evaluation 15,363 customers
The policy was locked on validation data before independent final evaluation.
Source: Plotly + Quarto interactive report · This is the static print version.

Reading guide

Repurchase diagnosis, the three-arm experiment, and customer-level policy evaluation.

1. CRM eligibility and exclusions

2. Cumulative repurchase by assignment month

3. Repurchase timing by customer group

4. Customer-group differences and next hypotheses

5. Three-arm assignment and baseline features

6. Incremental contribution profit per customer

7. Added coupon value and cost reconciliation

8. Side effects and operating conditions

9. Customer-level models and policy lock

10. Independent incremental profit of the locked policy

11. Constrained allocation and selection rationale

12. Side-effect upper bounds and final decision

13. Interpretation and scope

Report scope

Sections 1–4 diagnose natural repurchase; sections 5–8 report average training-experiment effects; sections 9–12 independently evaluate customer policies; section 13 covers interpretation and references.

All 13 sections, 17 tables, and 14 charts are included. PDF charts are vector images. Legend toggles and hover details described in the text are available in the original HTML.

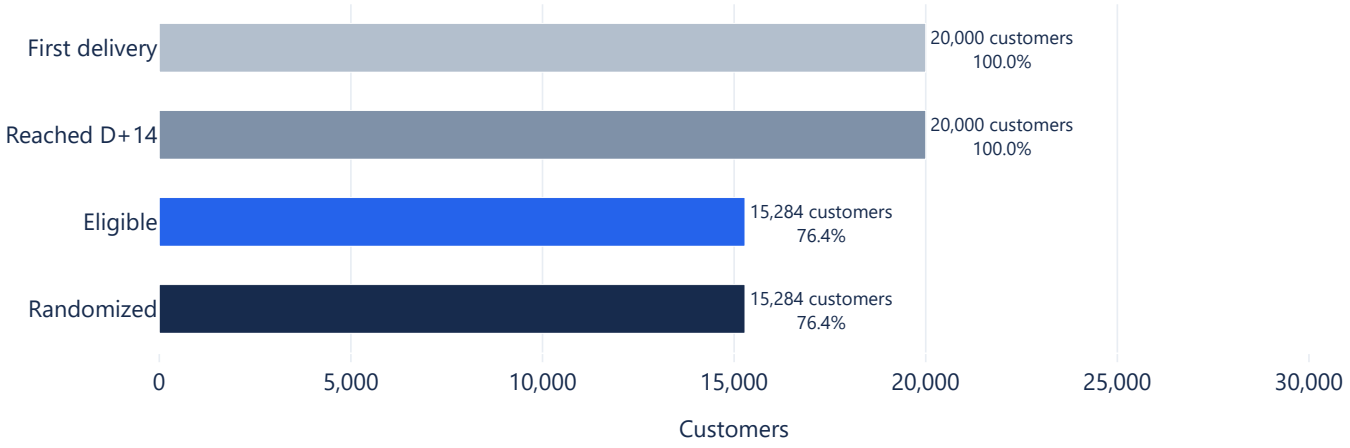
Read each table and chart with its stated window and denominator. Policy profit and side effects use the full final-evaluation population.

1. CRM eligibility and exclusions

Eligible customers made no additional payment during the 14 days after first delivery and meet first-order status, consent, and email requirements. Customers are randomized to no contact, information, or information plus a coupon. **The repurchase diagnosis in sections 1–4 uses only the no-contact arm.**

How many customers are eligible?

Synthetic data · Training · As of 2025-10-01
First-delivery customers: 20,000
Eligibility at the reference time
Snapshot sim-8aecf9722a5328b9



Eligibility flow, distinct from a churn rate.
Before D+14: 0 · Excluded: 4,716
Exclusion reasons may overlap

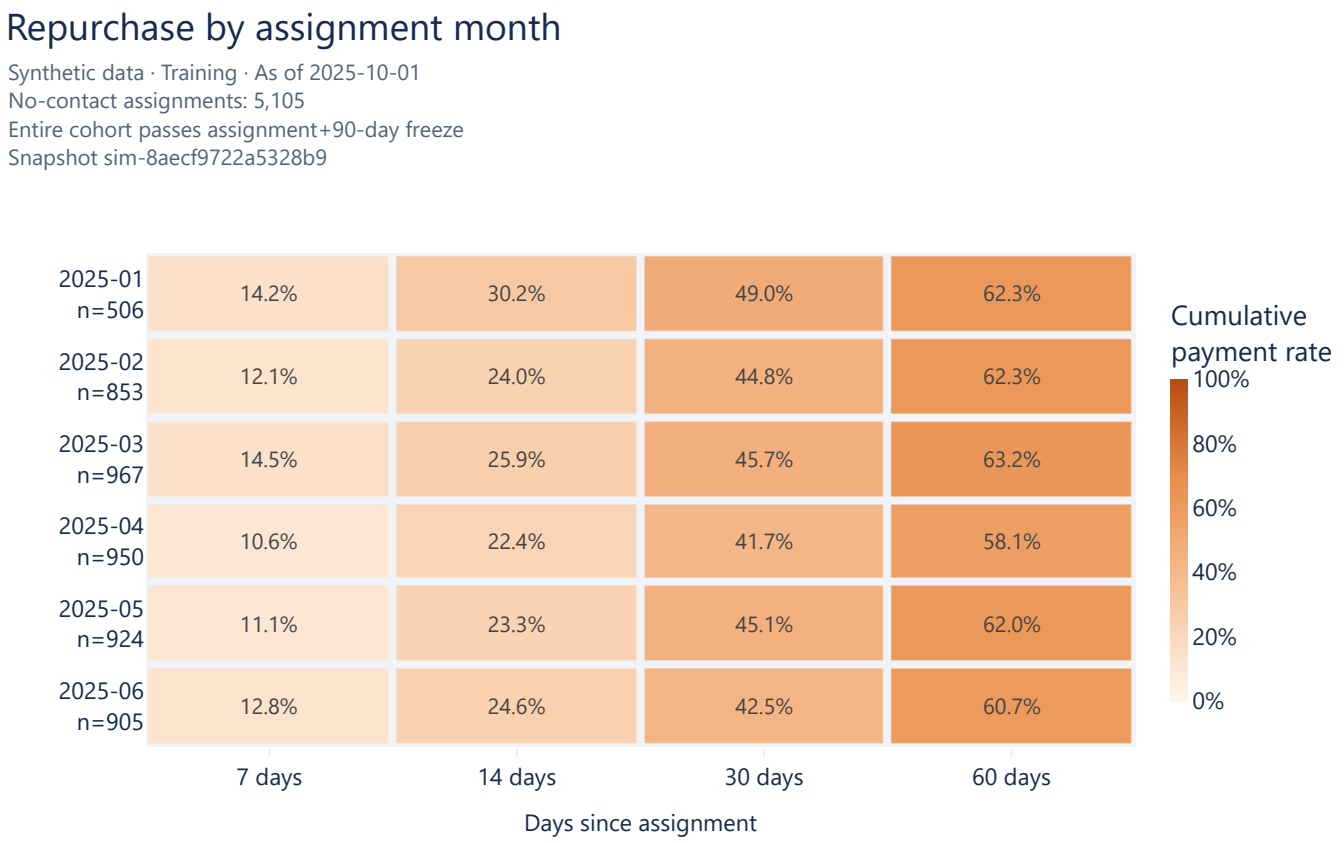
The 4,716 excluded customers include both customers who already repurchased and do not need this campaign and customers who do not meet contact requirements. They cannot all be interpreted as churned customers.

Exclusion reason	Customers · overlaps allowed
Additional payment before assignment	1,163
No email consent	2,873
Email unavailable	791
First order fully refunded	262

The denominator consists of customers with confirmed first delivery whose scheduled D+14 assignment falls in the training period. Exclusion reasons can overlap, so their counts may not sum to the number of excluded customers.

2. Cumulative repurchase by assignment month

Rows group no-contact customers by **assignment month**; columns show days **since assignment**. Darker cells indicate a higher proportion making an additional payment within that period. Hover for customer counts and 95% intervals.



NA: immature or no denominator; distinct from 0%
Entire cohort must pass the assignment+90-day freeze
Additional payments remain events after refunds

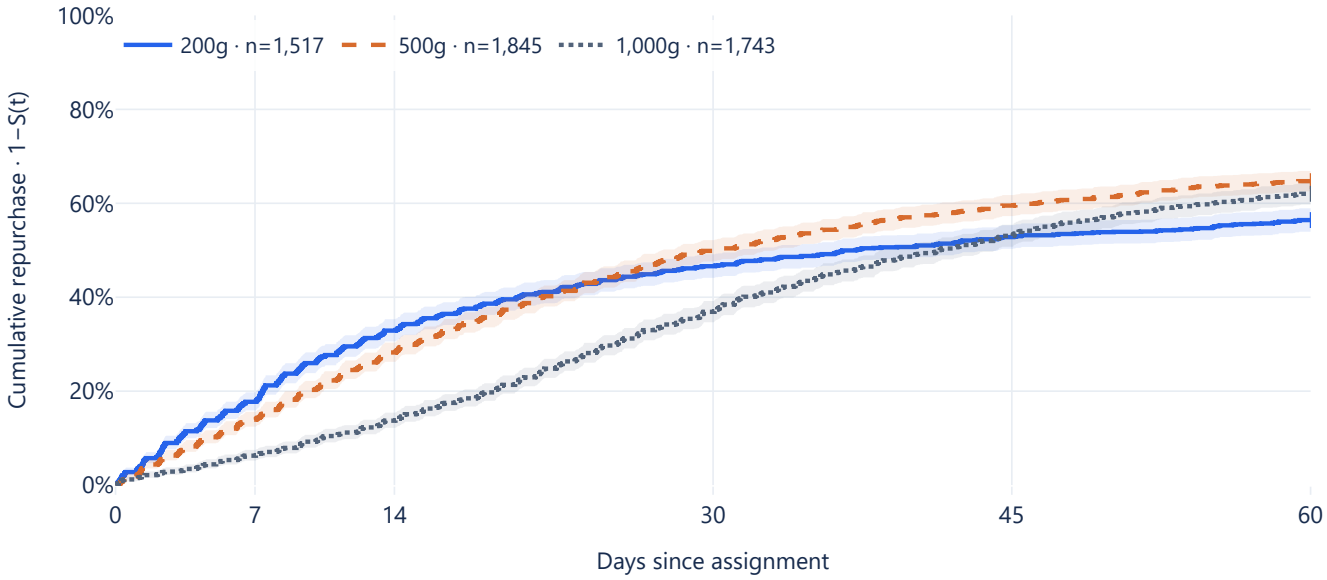
Monthly differences reflect both customer composition and the synthetic generation rules, so this table cannot establish that a particular month's operations or content were more effective. A row remains NA if the full monthly cohort has not completed 90-day settlement.

3. Repurchase timing by customer group

Cumulative repurchase in the full control group reaches 50% at 36.4 days after assignment, or approximately 50.4 days after first delivery. The curves show **Kaplan–Meier cumulative repurchase, 1−S(t)**, incorporating time to first repurchase. Customers without a purchase remain in the denominator while observed and are right-censored at observation end.

When do customers repurchase?

Synthetic data · Training · As of 2025-10-01
No-contact assignments: 5,105
Right-censored at observation end
Snapshot sim-8aecf9722a5328b9



KM 1−S(t) · Shading: pointwise 95% interval
Short ticks: censoring · Hover: risk-set size
Does not identify optimal timing or causal effects

Click a legend entry to toggle a package-size curve and its interval together. Hover over a curve to view the risk set and cumulative repurchase at that time.

Group	Before assignment	Before day 14	Before day 30	Before day 60
All	5,105	3,846	2,835	1,974
200g	1,517	1,018	810	661
500g	1,845	1,324	925	651
1,000g	1,743	1,504	1,100	662

The risk set comprises customers still observed who have not repurchased **immediately before** the stated time. Table values count customers. Customers whose follow-up ends at the 60-day boundary are included immediately before that boundary. Intervals are pointwise 95% Greenwood log-log intervals.

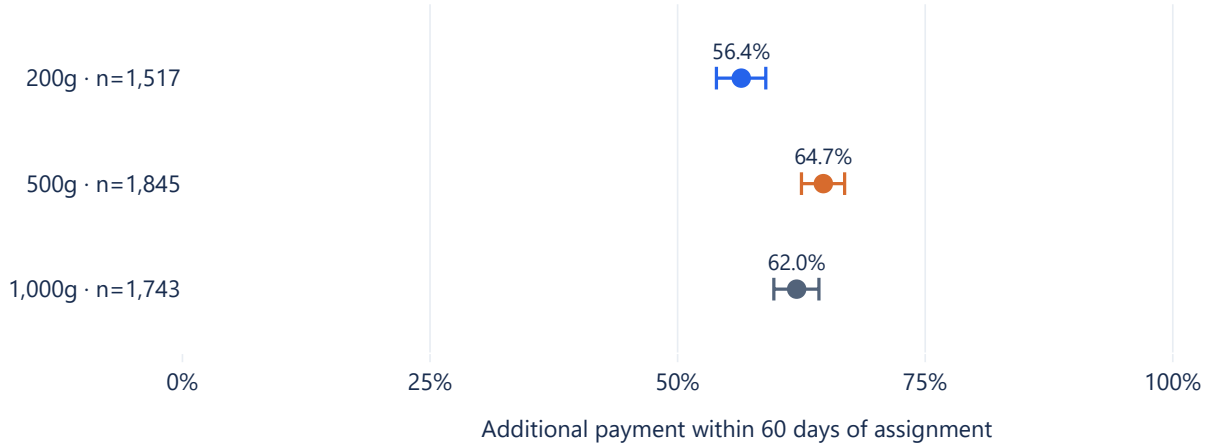
> **Planning implication:** natural repurchase timing can inform message scenarios. The sending day is fixed at D+14 in this experiment. A time read from these curves therefore cannot be established as the optimal sending day. Timing effects require a separate randomized experiment.

4. Customer-group differences and next hypotheses

60-day repurchase by first package: 500g: 64.7%, 200g: 56.4%. The observed gap of 8.3 pp describes a customer-group difference. It does not mean that changing package size or sending a coupon would produce that improvement.

Repurchase by package size

Synthetic data · Training · As of 2025-10-01
Settled control customers: 5,105
Labels frozen at assignment+90 days
Snapshot sim-8aecf9722a5328b9



Points: 60-day payment rate · Lines: Wilson 95% interval
n: customers with frozen labels · Descriptive size comparison
Customer-group differences are distinct from campaign effects

First package	Settled customers	Repurchasers	60-day repurchase	95% interval
200g	1,517	856	56.4%	53.9%–58.9%
500g	1,845	1,194	64.7%	62.5%–66.9%
1,000g	1,743	1,081	62.0%	59.7%–64.3%

Among the full control group of 5,105 customers, 3,131 repurchased, a rate of 61.3%, with a Wilson 95% interval of 60.0%–62.7%. First package size is observed before assignment, but may also reflect differences in purchase motivation, consumption, and other factors.

Diagnostic question	Next comparison	Decision metric
Does information help groups with low natural repurchase?	D+14 information versus no contact; prespecified exploration by package size	60-day incremental contribution profit and repurchase-rate difference
Are discounts reaching customers who would buy anyway?	Identical information plus coupon versus information alone	Incremental profit after coupon, shipping, and refund costs
Can targeting only low repurchase rates lose money?	Send-to-all, coupon-to-all, and customer-level policies	Validation incremental profit within budget; unsubscribe rate

Sections 5–8 report average training-experiment effects; sections 9–12 connect these to customer-level allocation. Models use training customers only. The policy was selected and locked on validation data before evaluating separately generated final customers.

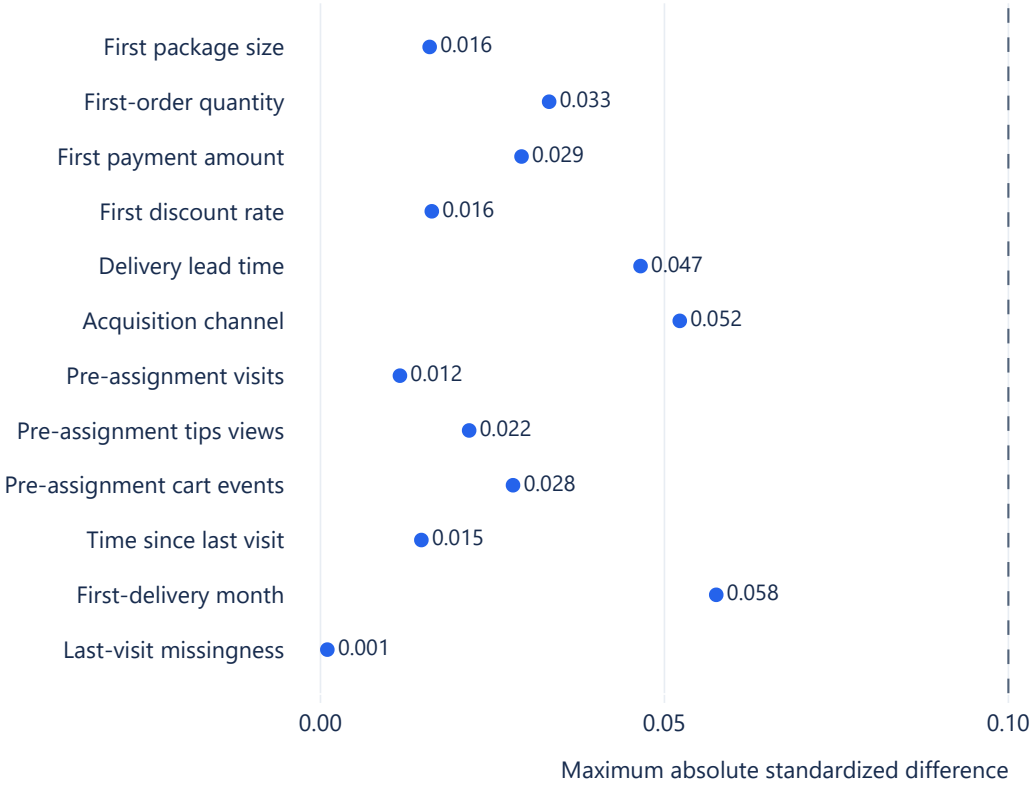
5. Three-arm assignment and baseline features

All **15,284 assigned training customers** completed the 60-day outcome window and 30-day settlement period. **ITT compares the full original assigned arms**, regardless of failed sends, unopened messages, or unused coupons. First-order profit is excluded.

Each arm was assigned with probability 1/3. The sample-ratio mismatch (SRM) test p-value is **0.8664**, above the investigation threshold of 0.001; no departure from the prespecified assignment ratio was detected. Independent reconciliation of the source ledger and marts matched customer-level accounts for all 15,284 customers.

Are baseline features balanced?

Synthetic data · Training · As of 2025-10-01
All ITT assignments: 15,284
Baseline features · All assigned customers
Snapshot sim-8aecf9722a5328b9



Points: maximum |SMD| across three comparisons
0.1 is a balance reference, not a causal validity threshold
Undefined values are shown as NA

The maximum absolute standardized mean difference (|SMD|) across the three pairwise comparisons of displayed baseline features is **0.058**; all displayed features are below the 0.1 reference. This diagnoses balance in observed features; it does not prove that randomization is perfect. Missingness in pre-assignment visit records was checked separately.

Message delivery and usage

Assigned arm	Assigned	Attempted	Reached	Opened	Redeemed
No contact	5,105	Not applicable	Not applicable	Not applicable	Not applicable
Information	5,064	5,064	4,757	2,066	Not applicable
Information + coupon	5,115	5,115	4,808	1,980	837

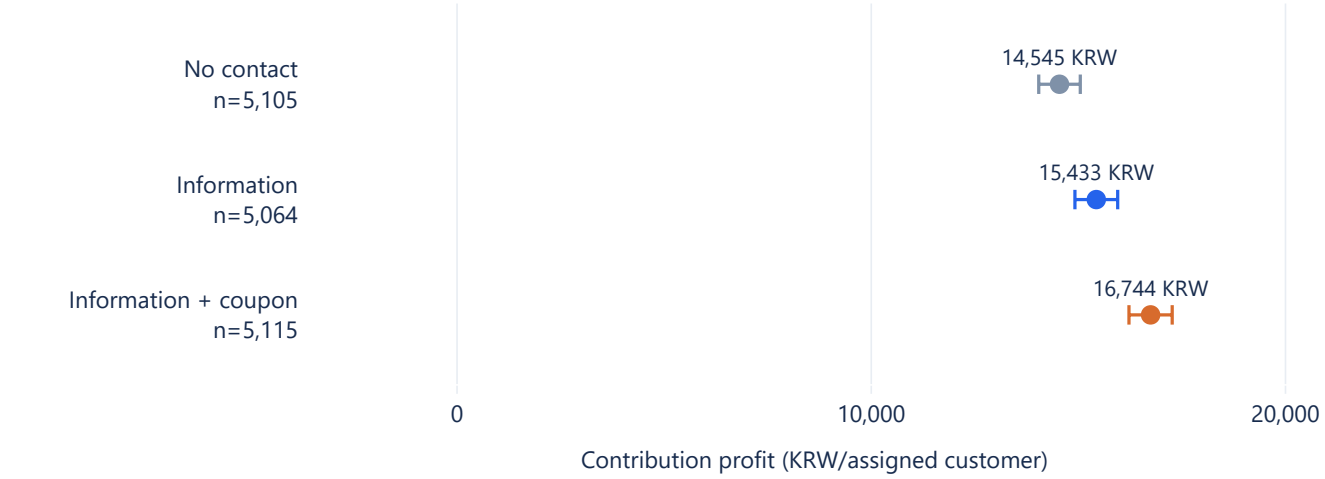
Each cell counts customers. Unreached customers remain in the profit-comparison denominator. Opens and coupon redemption do not necessarily form a sequential funnel; effects are not calculated only among openers or redeemers.

6. Incremental contribution profit per customer

Each arm's 60-day contribution profit accounts for discounts, refunds, product costs, shipping, payment fees, and sending costs. Averages below are in **KRW per assigned customer**, distinct from total revenue or averages among purchasers only.

Contribution profit per customer

Synthetic data · Training · As of 2025-10-01
All ITT assignments: 15,284
60-day outcomes + 30-day settlement complete
Snapshot sim-8aecf9722a5328b9



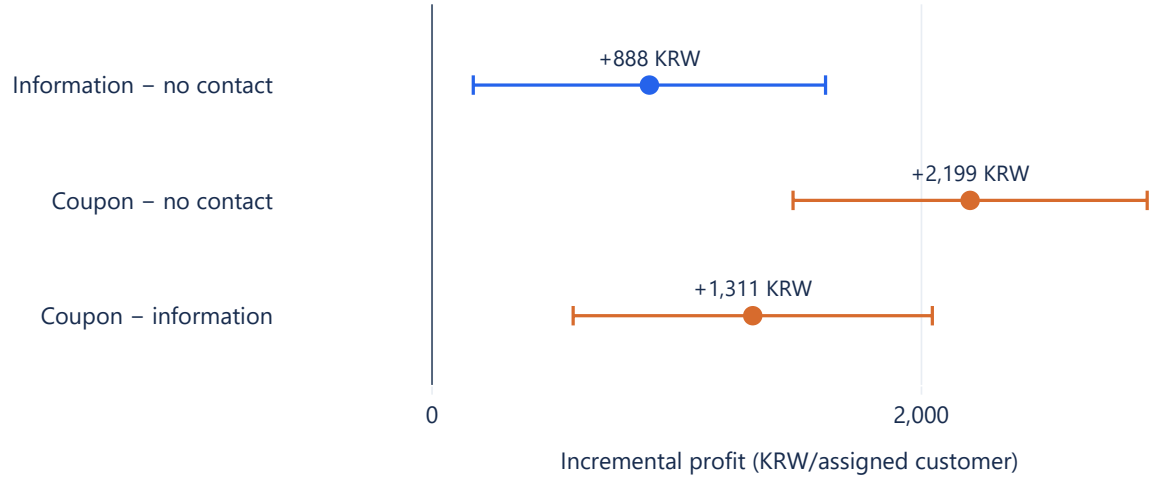
Points: arm means · Lines: individual 95% intervals
Denominator: all originally assigned customers
Includes failed sends and unopened messages

Assigned arm	ITT customers	Mean contribution profit	95% interval for mean	Repurchase rate
No contact	5,105	14,545 KRW	14,043~15,046 KRW	61.3%
Information	5,064	15,433 KRW	14,917~15,949 KRW	62.5%
Information + coupon	5,115	16,744 KRW	16,222~17,265 KRW	67.8%

Incremental profit from information and coupons

How much incremental profit?

Synthetic data · Training · As of 2025-10-01
All ITT assignments: 15,284
60-day outcomes + 30-day settlement complete
Snapshot sim-8aecf9722a5328b9



Lines: individual 95% intervals, not simultaneous
Hover for Holm-adjusted p-values
Coupon arm adds a coupon to identical information

Comparison	Incremental profit / assigned customer	Individual 95% interval	Holm-adjusted p	Interpretation within synthetic world
Information − no contact	+888 KRW	+169~+1,608 KRW	0.0156	Evidence of an increase
Information + coupon − no contact	+2,199 KRW	+1,475~+2,922 KRW	<0.001	Evidence of an increase
Information + coupon − information	+1,311 KRW	+577~+2,044 KRW	<0.001	Evidence of an increase

Lines are Welch **individual 95% confidence intervals**, not simultaneous intervals for the three comparisons. Table p-values use Holm correction across the three prespecified contribution-profit comparisons.

Incremental profit for information + coupon versus no contact is **+2,199 KRW/customer**, and the difference versus the same information alone is **+1,311 KRW/customer**. The added value of the coupon must be judged by directly comparing the information + coupon and information-only arms.

Repurchase with the same ITT denominator

Comparison	Repurchase-rate difference	Individual 95% interval · exploratory
Information − no contact	+1.19 pp	-0.70 pp~+3.07 pp
Information + coupon − no contact	+6.45 pp	+4.60 pp~+8.30 pp
Information + coupon − information	+5.26 pp	+3.41 pp~+7.11 pp

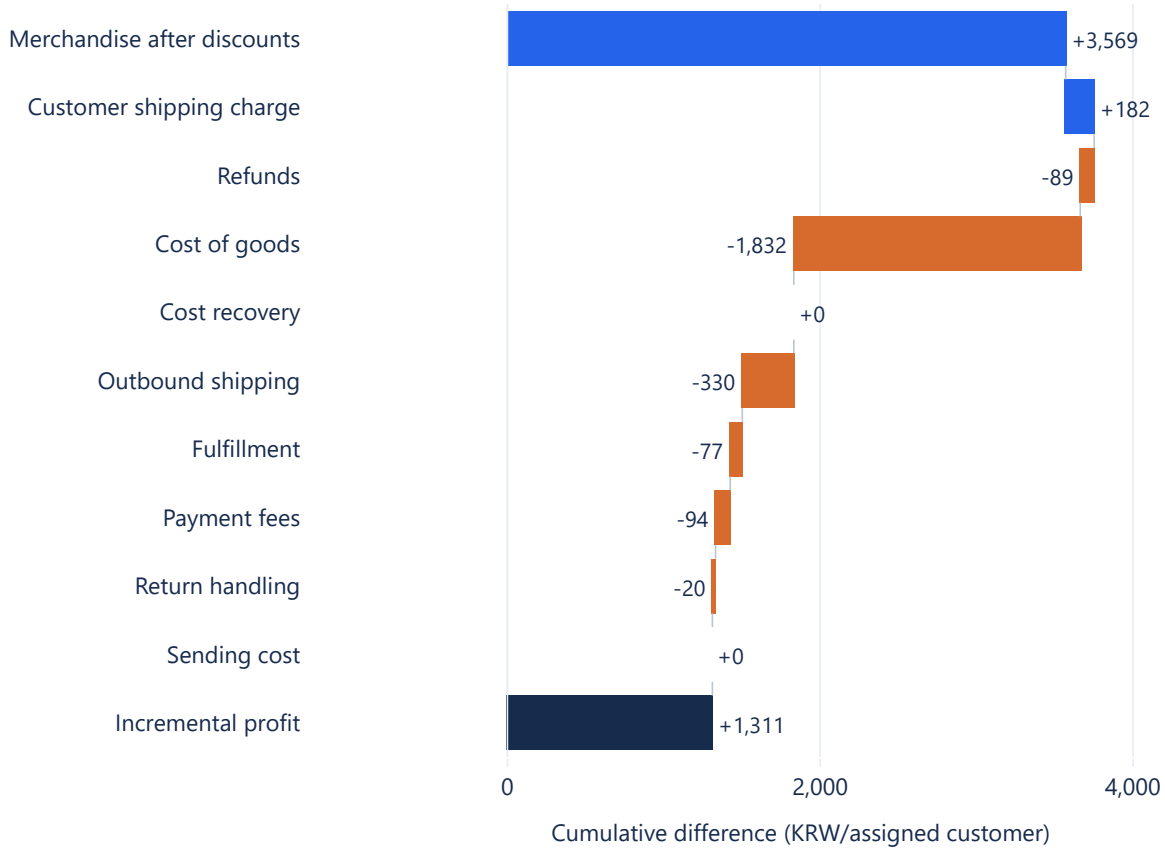
Rate-difference intervals use the Newcombe score method and are secondary analyses. A higher repurchase rate alone does not establish profitability; contribution profit above is the primary metric.

7. Added coupon value and cost reconciliation

The waterfall sums per-customer accounting differences for **information + coupon − information**. Positive components increase the profit difference and negative components reduce it. The final total matches the experiment estimate of +1,311 KRW/customer above.

Components of added coupon profit

Synthetic data · Training · As of 2025-10-01
All ITT assignments: 15,284
60-day outcomes + 30-day settlement complete
Snapshot sim-8aecf9722a5328b9



Information + coupon − information · Per-customer means
Discounts already enter revenue; no duplicate deduction
Accounting decomposition, not component causal effects

Coupon discounts already enter the waterfall's merchandise revenue after discounts, so coupon cost is not deducted again. This is an accounting decomposition; components are not interpreted as independent causal mechanisms or mediation effects. The total profit-difference interval is reported above instead of component-level uncertainty.

Among the coupon arm's 5,115 customers, 837 redeemed a coupon; actual CRM coupon discounts average **491 KRW per assigned customer**. Redeemed discounts differ from the KRW 3,000 budget reserved at issuance.

Assigned arm	List-price merchandise	Other discounts	CRM coupon discounts	Revenue after discounts	Sending cost
No contact	34,133 KRW	0 KRW	0 KRW	34,133 KRW	0 KRW
Information	36,114 KRW	0 KRW	0 KRW	36,114 KRW	20 KRW
Information + coupon	40,174 KRW	0 KRW	491 KRW	39,683 KRW	20 KRW

All table amounts are per assigned customer, with discounts and sending costs shown as positive expenditures. **List-price merchandise amount − other discounts − CRM coupon discounts = merchandise revenue after discounts.** Displayed amounts are rounded to whole KRW; calculations use unrounded means.

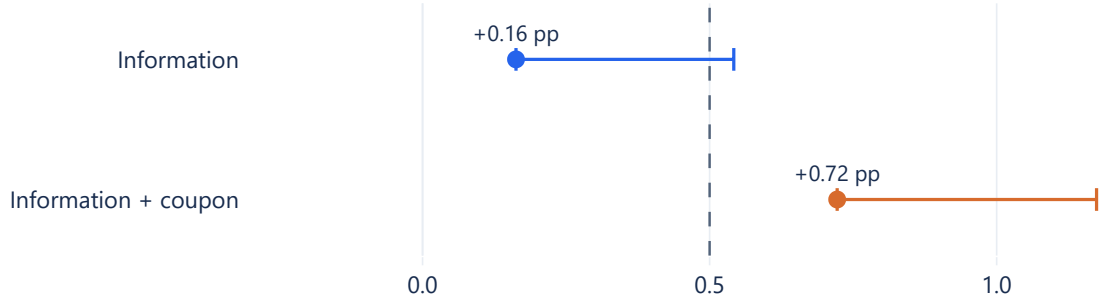
8. Side effects and operating conditions

Alongside profit, we assess increases in unsubscribe and refund-customer rates versus no contact. Tolerances are **+0.5 pp and +1.0 pp**, respectively. Decisions compare the uncertainty-adjusted upper bound with tolerance, rather than relying on the point estimate alone.

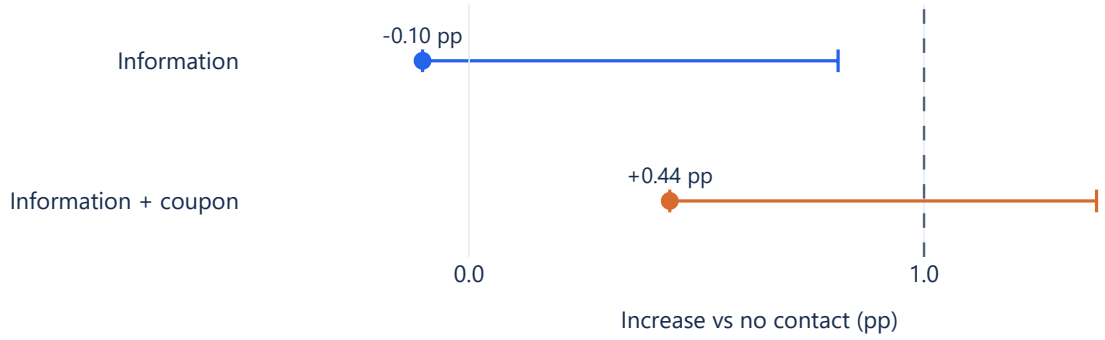
Are side effects within tolerance?

Synthetic data · Training · As of 2025-10-01
All ITT assignments: 15,284
Versus no contact · Correction across 4 upper bounds
Snapshot sim-8aecf9722a5328b9

Unsubscribe · Tolerance 0.5 pp



Refund customers · Tolerance 1 pp



Point: risk difference · Right end: adjusted one-sided bound
Dashed line: prespecified tolerance; compare the upper bound
Safety assessment differs from two-sided 95% intervals

Versus no contact	Metric	Observed increase	Adjusted one-sided upper bound	Tolerance	Assessment
Information	Unsubscribe	+0.16 pp	+0.54 pp	+0.50 pp	Not established
Information	Refund-customer rate	-0.10 pp	+0.81 pp	+1.00 pp	Within tolerance
Information + coupon	Unsubscribe	+0.72 pp	+1.17 pp	+0.50 pp	Not established
Information + coupon	Refund-customer rate	+0.44 pp	+1.38 pp	+1.00 pp	Not established

A one-sided family alpha of 0.05 is divided across **4 comparisons** (two active arms × two metrics), giving **98.75% one-sided upper bounds**. Chart lines run from the estimate to the upper bound and differ from two-sided 95% intervals. “Not established” means the upper bound is not within tolerance; it does not establish that deterioration necessarily exceeds tolerance.

Planning decision at this stage

Option	Contribution profit vs no contact	Side-effect conditions	Planning decision
Information	Evidence of an increase	Unsubscribe: tolerance not established	Modify operations and validate further
Information + coupon	Evidence of an increase	Unsubscribe·Refund-customer rate: tolerance not established	Modify operations and validate further

> The design caps contact at 30% of eligible customers and reserves up to KRW 3,000,000 in variable costs per 10,000 customers. **Profit, side effects, and operating constraints must be checked together.** These three-arm averages are not final customer-policy results, so an arm that sends everyone the same message is not approved as a final campaign satisfying those constraints.

In the following policy stage, these results serve as baselines for selecting customers and benefits using pre-assignment features. Changes to content, frequency, or subscription preferences that could reduce contact burden are separate experimental hypotheses. Proposing real operational expansion requires independent evaluation of a locked policy and confirmation of side-effect conditions. Validation and final-evaluation data were not used in these arm comparisons.

Experiment calculation and reproduction

Inputs are the same synthetic source as the diagnosis. Results are in `report/data/experiment.json`; assignment, balance, delivery, and accounting checks are in `report/data/experiment_checks.json`. The `compare-arms` command regenerates diagnostic and experiment charts together, and Quarto bundles them into one HTML file.

Mean inference refers to SciPy Welch t-test; independent proportion-difference intervals refer to the Newcombe implementation. Independent pilot power simulation has not been performed. Sample size and seeds were not changed in response to these results.

9. Customer-level models and policy lock

Building on the three-arm average comparisons, 11 pre-assignment features predict **conditional contribution profit and side-effect probabilities** for no contact, information, and information plus coupon. Only the assigned treatment's outcome is observed for each customer. Model differences therefore estimate **conditional average effects**, not observed individual true effects.

Purpose	Assignment dates	All assigned customers	Prespecified label cutoff
Training	2025-01-15 inclusive–2025-07-01 exclusive	15,284	2025-10-01
Validation / selection	2025-10-01 inclusive–2025-12-01 exclusive	15,167	2026-03-01
Locked-policy final evaluation	2026-03-01 inclusive–2026-05-01 exclusive	15,363	2026-08-01

Each split retains the full originally randomized population with completed 90-day settlement. Preprocessing and models are fitted on training data only; validation selects candidates, and the final split evaluates the locked policy only. Models, λ, and policies are not reselected after seeing final results.

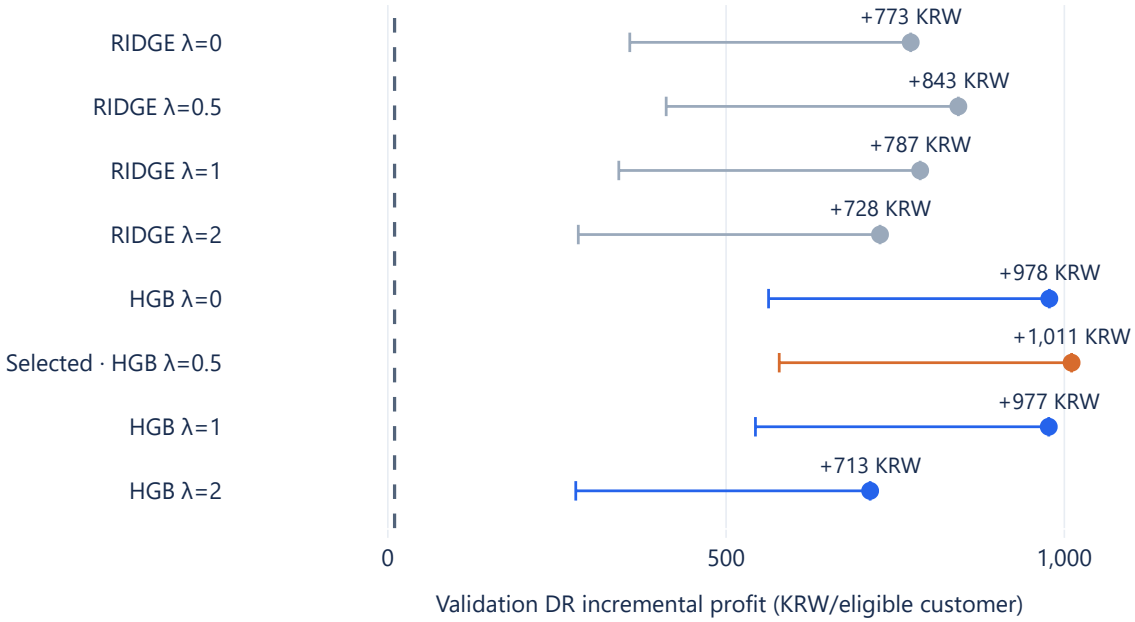
Two models and eight policy candidates

The training set contains **15,284 customers**. Candidates are Ridge(α=100) and histogram gradient boosting (HGB). A T-learner fits separate contribution-profit models for each arm; unsubscribe and refund outcomes are modeled as classification probabilities. Negative and 0 profit values are retained. Numeric imputation, scaling, and categorical encoding are fitted on training data only. Months/channels first appearing later are handled as unseen categories.

Applying reservation-cost penalties λ=0, 0.5, 1, 2 to each model gives 8 candidates for validation. Sending candidates are ranked by **predicted incremental profit − λ×reserved cost**. The λ penalty allocates scarce resources; it is not deducted as an accounting cost from evaluated contribution profit. All policies share prespecified HGB outcome predictions for DR evaluation, separating policy ranking models from the evaluation nuisance model.

Which policy was selected?

Synthetic data · Validation · Candidate selection
As of 2026-03-01 · Eligible customers: 15,167
Snapshot sim-35f0627d627d5ab7



Point: validation DR · Left end: selection lower bound
Orange: selected · Blue: feasible · Gray: infeasible
Used for candidate selection; final performance is separate

Candidate	DR incremental profit / customer	Validation approximate 95% interval	Profit / side effects	Selection
Ridge · λ=0	+773 KRW	+358~+1,188	Pass / Fail	—
Ridge · λ=0.5	+843 KRW	+411~+1,275	Pass / Fail	—
Ridge · λ=1	+787 KRW	+341~+1,232	Pass / Fail	—
Ridge · λ=2	+728 KRW	+282~+1,174	Pass / Fail	—
HGB · λ=0	+978 KRW	+563~+1,393	Pass / Pass	—
HGB · λ=0.5	+1,011 KRW	+579~+1,443	Pass / Pass	Locked
HGB · λ=1	+977 KRW	+543~+1,411	Pass / Pass	—
HGB · λ=2	+713 KRW	+278~+1,148	Pass / Pass	—

The profit criterion requires the lower bound of the validation DR approximate 95% interval to exceed fixed setup cost per customer. Side-effect screening covers 8 candidates × 2 metrics and uses Bonferroni-corrected **99.6875% one-sided upper bounds**. Validation intervals are normal approximations for fixed action scores; they are not final inference accounting for all allocation dependence and selection optimism.

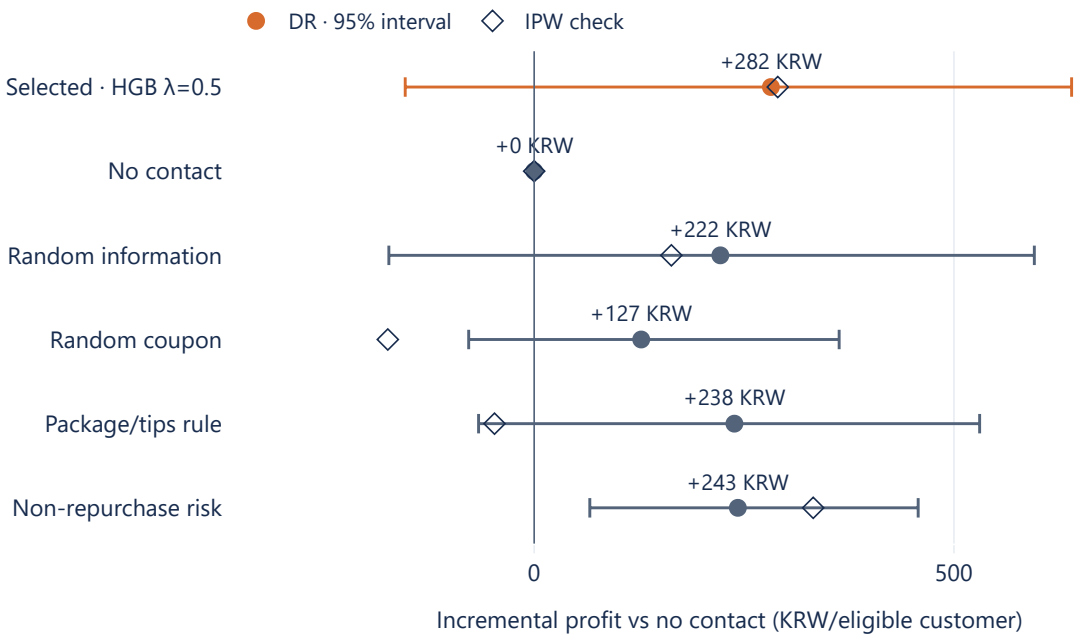
Among feasible candidates, prefer the simpler Ridge model if it lies within 1 standard error of the highest DR estimate. Within a model, rank by decreasing DR, increasing λ, then fixed identifier. Select no contact if no candidate is feasible. Validation yielded 4 feasible candidates, including 3 within 1 standard error. **HGB · λ=0.5** was selected and locked.

10. Independent incremental profit of the locked policy

The final denominator is **all 15,363 assigned customers**; the analysis does not retain only customers whose assigned arm matches the policy action. Actual randomization probabilities compare policy value with the value of no contact for everyone. DR combines outcome predictions with assignment probabilities as the primary analysis; IPW provides a probability-weighted check.

Does the locked policy add profit?

Synthetic data · Final evaluation · After policy lock
As of 2026-08-01 · Eligible customers: 15,363
Snapshot sim-4c3a39106c7ee7c7



DR: primary estimator · IPW: check estimator
Differences already include actual coupon and sending costs
Intervals do not establish multiple-baseline superiority

Policy	DR incremental profit / customer	95% interval · KRW	IPW check / customer
Locked policy: HGB · λ=0.5	+282 KRW	-154~+640	+290 KRW
Keep no contact	+0 KRW	+0~+0	+0 KRW
Random information priority	+222 KRW	-173~+596	+163 KRW
Random coupon priority	+127 KRW	-78~+363	-174 KRW
Package/tips rule	+238 KRW	-66~+531	-47 KRW
Non-repurchase risk priority	+243 KRW	+66~+457	+332 KRW

The locked policy's primary estimate is **+282 KRW/eligible customer**, with a 95% interval of **-154~+640 KRW**. The IPW check is +290 KRW, with an interval of -213~+798 KRW. Other rows are prespecified **exploratory baselines**. They are not used to choose a new final winner, nor to claim simultaneous superiority over multiple baselines.

For the same locked policy, DR incremental profit moved from +1,010.73 KRW/customer on validation to +281.94 KRW/customer on final data; the estimate decreased. Sampling variation, validation selection optimism, and customer-composition differences are possible explanations, but this comparison cannot separate or establish the cause.

Customers are resampled with replacement within assignment days for **2,000 bootstrap repetitions**. Repeated customers occupy separate slots; allocation and constraints are recalculated for the same policy in each repetition, without model refitting or policy reselection. The intervals therefore condition on the locked models/policy and observed daily composition, excluding full training uncertainty and uncertainty about generalization to real markets.

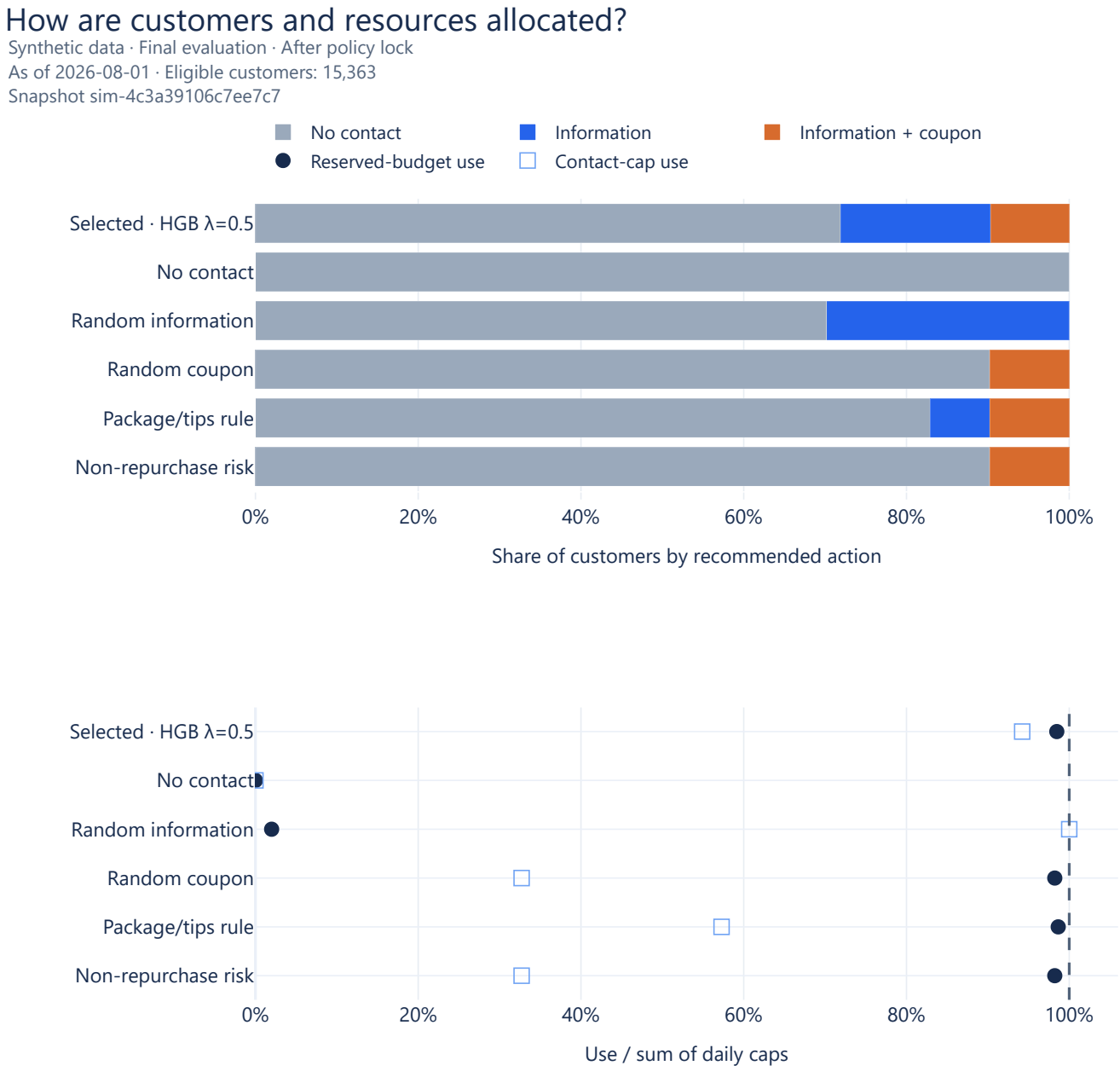
Scaling to rollout size and accounting for costs

For **10,000 eligible customers**, scaled incremental profit after setup cost is **+2,719,396 KRW**, with an interval of **-1,636,266~+6,301,281 KRW**. The calculation is `DR incremental profit per customer × rollout population − fixed setup cost`; setup cost here is 100,000 KRW. Setup cost is charged only when at least one customer is contacted; the bootstrap reapplies this condition.

Discounts, refunds, product costs, shipping, payment, and sending costs already enter customer-level contribution profit. Reservation budgets and λ penalties are not deducted again. Scaling is a planning calculation conditional on unchanged customer composition and operating conditions, not a forecast of actual sales.

11. Constrained allocation and selection rationale

The contact cap is **30%** of eligible customers; the variable-cost reservation budget is **3,000,000 KRW per 10,000 customers**. Each assignment day's caps are proportional to its eligible population and rounded down to integers. Coupons reserve **full face value plus sending cost**, without discounting by expected redemption. Fixed setup cost of 100,000 KRW is accounted for separately, outside the variable-cost reservation cap.



Top: allocation · Bottom: budget/contact-cap use
Coupon face value is fully reserved regardless of redemption
Daily batch caps are checked separately from totals

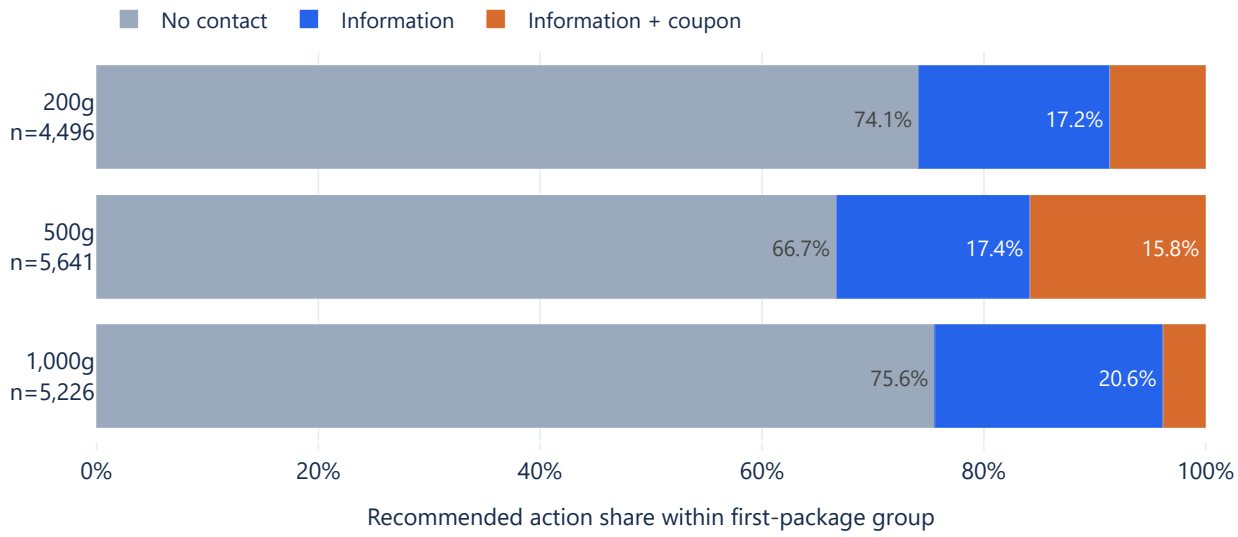
Final population	No contact	Information	Information + coupon
15,363	11,045	2,834	1,484

The evaluation population receives 4,318 contacts (28.1%), against a total contact cap of 4,583. Reserved cost is **4,538,360 KRW**, against summed daily caps of 4,608,900 KRW. The original allocation has 0 violating days and the bootstrap has 0 violating repetitions: **All assignment-day constraints met**.

Actions with positive predicted incremental profit are allocated by score, with at most one action per customer. Only positive predicted increases in unsubscribe/refund risk are accumulated, so one customer's predicted risk reduction cannot offset another's predicted increase. This allocation filter does not replace the observed-outcome side-effect evaluation in the next section. The greedy implementation respects constraints but **does not guarantee a global optimum**.

Recommended actions by package

Synthetic data · Final evaluation · After policy lock
As of 2026-08-01 · Eligible customers: 15,363
Snapshot sim-4c3a39106c7ee7c7



First package size is a pre-assignment feature
Recommendations, distinct from actual sending performance
Does not establish true individual effects

First package	Customers	No contact	Information	Information + coupon
200g	4,496	3,332	775	389
500g	5,641	3,763	984	894
1,000g	5,226	3,950	1,075	201

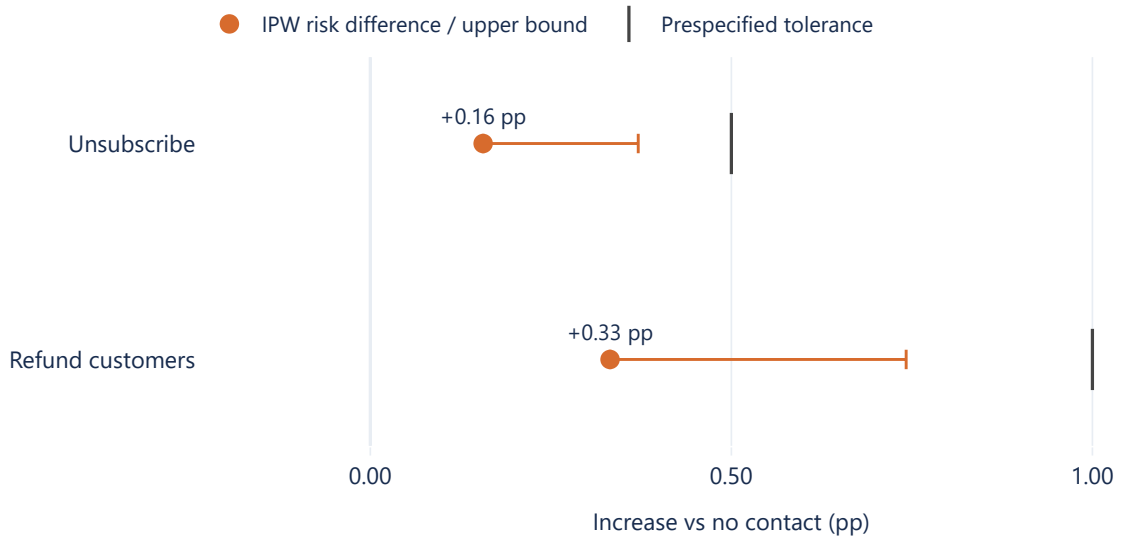
The first-package table explains whom the locked policy selects. It does not establish true individual or segment-level causal effects. The non-repurchase risk baseline ranks the chance of not repurchasing, not the incremental response to a message. Random, rule, and risk baselines use the same contact, reservation-budget, predicted-risk filters, and positive incremental-profit requirement.

12. Side-effect upper bounds and final decision

The locked policy's unsubscribe and refund-customer rates are compared with no contact for everyone. Differences below are IPW estimates from observed labels, distinct from model-predicted risk scores.

Side effects of the selected policy

Synthetic data · Final evaluation · After policy lock
As of 2026-08-01 · Eligible customers: 15,363
Snapshot sim-4c3a39106c7ee7c7



Full-policy IPW risk difference · Two-metric correction
Compare each right-end upper bound with its tolerance
Not a simple rate among selected customers only

Locked policy – no contact	IPW increase	One-sided upper bound	Tolerance	Assessment
Unsubscribe	+0.16 pp	+0.37 pp	+0.50 pp	Within tolerance
Refund-customer rate	+0.33 pp	+0.74 pp	+1.00 pp	Within tolerance

A one-sided family α=0.05 across two metrics yields **97.50% one-sided upper bounds**. Their scope differs from the 4 comparisons in the arm-average analysis and from validation candidate screening. Intervals for multiple exploratory baselines do not provide simultaneous safety guarantees.

The number of IPW customers whose assigned arm matches the policy action is 5,195, with an effective sample size of 5,195.0. Matched counts and effective sample size diagnose weighted information; the analysis denominator remains 15,363 customers. Dependence on matched counts and the DR/IPW difference are assessed together.

> **Modify operating conditions or collect more evidence.** Evidence is not yet sufficient to jointly establish the profit lower-bound, both side-effect, and resource criteria.

The policy is not reoptimized on the same final split after observing its result. Revisions require a new experiment with separate customers and periods.

Methods, records, and remaining validation

Public aggregates are in `report/data/policy.json`; prespecified rules are in `docs/STAGE_4_POLICY_SPEC.md`. The actual UTC execution time of policy locking differs from the synthetic business dates in the tables. File creation order is also checked to confirm that the lock precedes final-source generation. Independent pilot power simulations and repeated inference/false-expansion checks in A/A null-effect worlds remain incomplete.

Models refer to scikit-learn Ridge and HistGradientBoostingRegressor; DR policy evaluation refers to Dudík·Langford·Li, Doubly Robust Policy Evaluation and Learning.

13. Interpretation and scope

- **Synthetic world:** data follow prespecified generation assumptions, not actual consumer research or real-company/brand performance records. Statistical significance concerns mean differences within this fixed synthetic world and does not validate real-brand performance. The 95% intervals describe sampling uncertainty in that world, not the probability of success in a real market. Policy evaluation uses a wholly fictional RCT and does not authorize immediate sending for a real brand.
- **Population:** each analysis split includes eligible customers who have not repurchased by D+14. Results are not generalized to all first-time buyers or customers without consent.
- **Feature timing:** both event time and recorded time must precede assignment. Arm labels are metadata and are excluded from prediction features.
- **Outcome definition:** an additional payment remains a repurchase event even if later refunded. Contribution profit separately accounts for refunds, shipping, payment, and sending costs.
- **Window boundaries:** payments are counted in `[assignment, assignment+60 days)`; final records must precede assignment+90 days. Unsettled outcomes are not replaced with 0.
- **Analysis separation:** hidden generator traits (oracle) were not used. Sections 1–8 use training customers only, excluding validation and final customers.

Reproduction and method references

The HTML version combines Plotly charts in Quarto as a single file containing scripts and styles; chart display works without an external CDN. Original aggregates are in `report/data/diagnostics.json`, input/code hashes in `report/data/analysis_manifest.json`, and mart-building code in `src/crm_retention_lab/marts.py`.

Statistical methods refer to lifelines KaplanMeierFitter; output construction refers to Plotly HTML export and Quarto HTML documents. Implemented survival probabilities and intervals are independently cross-checked against lifelines.